Table 1: The 12 systems. Design decisions use dev systems only; test systems are run once with the frozen recipe. Model error: median $\|f_{\text{real}} - f_{\text{model}}\|/\|f_{\text{real}}\|$ over a Sobol sample of the IC box and the input box, for the partial-knowledge model (nominal parameters) and for a 30% parameter error (full model).

system	role	n_x	n_y	n_z	t_{ref}	input enters	unmodelled term (partial model)	err. partial	err. 30%
battery	dev	5	2	7	1	current (SOC + 2-RC + thermal; 1-min time unit; V_m and thermistor measured)	the slow RC branch $R_2 \parallel C_2$ (model: the textbook 1-RC equivalent circuit; V_2 is not driven, not in the terminal voltage and dissipates no heat)	0.22	0.09
bioreactor	dev	2	1	5	1	bilinear (dilution rate)	Monod saturation of the growth rate (model: first-order kinetics $\mu_{\max} s/K_s$)	2.15	0.50
duffing	dev	2	1	5	1	state-dependent gain	the cubic stiffness $-x_1^3$ (model: linear spring)	0.13	0.02
lorenz	dev	3	1	7	1	additive, chaotic	the coupling $-x_1 x_3$ in the measured equation	0.71	0.29
quad_tank	dev	4	2	6	15	distributed, non-minimum phase	the square-root orifice law (model: flows linearised at the nominal equilibrium)	0.15	0.69
vdp	dev	2	1	5	1	additive + saturating actuator	the nonlinear damping law $\mu(1 - x_1^2)x_2$ (model: undamped oscillator)	0.66	0.20
cartpole	test	4	2	6	1	matrix-valued $G(x)$, balancing loop	large-angle pendulum terms (model: small-angle $\sin\theta = \theta$, $\cos\theta = 1$, no ω^2)	0.03	0.35
fhn	test	2	1	5	2	additive	the recovery leak $-\varepsilon b w$	0.04	0.16
highway_traffic	test	5	3	12	10	ramp inflow	congestion in the fundamental diagram (model: free-flow $q = v_{\text{free}}\rho$)	0.82	0.40
network	test	10	3	12	1.5	node 0 of a ring of 5 FHN neurons (nodes 0, 1, 3 measured)	the coupling edge 2-3 of the ring (model: chain 3-4-0-1-2, graph partly unknown)	0.22	0.21
rossler	test	3	1	7	8	additive, chaotic	the folding nonlinearity $x_1 x_3$	0.30	0.25
two_link_arm	test	4	2	6	2	$M(q)^{-1} e_1$	Coriolis and centrifugal terms (model: $M(q)\ddot{q} = \tau - g(q) - k_d\dot{q}$)	0.04	0.29